

9. Vectors $\mathbf{u}$ and $\mathbf{v}$ have components $\begin{pmatrix} p \\ -2 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 2p+16 \\ -3 \\ 6 \end{pmatrix}$, $p \in \mathbb{R}$.

(a) (i) Find an expression for $\mathbf{u} \cdot \mathbf{v}$.

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(ii) Determine the values of p for which $\mathbf{u}$ and $\mathbf{v}$ are perpendicular.

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(b) Determine the value of p for which $\mathbf{u}$ and $\mathbf{v}$ are parallel.

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